

Structure-informed deep generation enables de novo metabolite annotation in untargeted metabolomics

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Metabolite annotation, especially the discovery of unknown metabolites, remains a fundamental challenge in mass spectrometry-based untargeted metabolomics due to limited reference mass spectra. Here we present MetGenX, a structure-informed encoder-decoder neural network that enables efficient and controllable generation of metabolite structures directly from MS2 spectra. By reformulating the spectrum-to-structure task as a structure-to-structure generation problem, MetGenX significantly improves generation accuracy and chemical space coverage. In independent tests, it achieved top-1 accuracy of 55.9% on 1388 NIST MS2 spectra and 68.5% on 1681 spectra from real biological samples, outperforming existing in silico tools. Its structure-informed design ensures robust performance across both positive and negative ionization modes without retraining. Applying a multi-step annotation workflow to mouse liver untargeted metabolomics data, MetGenX identified two previously uncharacterized metabolites absent from major human metabolome databases. These results demonstrate MetGenX's strong potential to advance de novo metabolite annotation and facilitate the discovery of uncharacterized chemical entities.

Untargeted metabolomics enables comprehensive and quantitative profiling of metabolites in biological systems. However, despite significant advances in instrumentation, accurate metabolite annotation remains a major bottleneck, particularly in liquid chromatography–mass spectrometry (LC–MS)-based workflows^{1,2}. Library-based spectral matching remains the gold standard for annotation but is limited to known metabolites with available reference spectra^{3,4}. The annotation of unknown metabolites, including “known unknowns” (metabolites present in structural databases but lacking reference spectra) and “unknown unknowns” (metabolites with previously uncharacterized structures), continues to pose a fundamental challenge⁵. To overcome this, computational approaches such as molecular networking and in silico prediction have emerged as powerful and complementary strategies for annotating unknown metabolites^{6,7}.

Molecular networking approaches cluster experimental features based on MS2 spectral similarity and leverage fragment-based substructure information from annotated neighbors to facilitate the discovery of unknown metabolites^{8,9}. These known structures provide essential context for inferring related unknowns, enabling more informed and targeted annotation strategies. Using this approach, Quinn et. al. discovered a series of amino acid conjugations of host bile acids, which were enriched in patients with inflammatory bowel disease or cystic fibrosis¹⁰. Alternatively, in silico prediction approaches—primarily based on machine learning and deep-learning—learn fragmentation patterns and rules from available reference spectra of known metabolites. These methods enable either the forward prediction of in silico MS2 spectra from molecular structures (e.g., CFM-ID¹¹ and MS-FINDER¹²) or the inverse inference of structural representations, such as molecular fingerprints, directly from experimental

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MS2 spectra (e.g., CSI:FingerID¹³ and MIST¹⁴). However, most in silico tools rely heavily on existing structural databases, making it difficult to identify metabolites with previously uncharacterized structures that are absent from known chemical space. Furthermore, the effectiveness of in silico prediction is currently constrained by the limited availability of high-quality reference MS2 spectra¹⁵. As a result, the full potential of in silico prediction for metabolite annotation has yet to be fully realized.

Recent advances in deep generation have enabled the discovery of chemical structures beyond known chemical space, with growing applications in computational drug design¹⁶. Various models have been developed to generate molecules, which can be optimized for specific targets using transfer learning, contrastive fine-tuning, and reinforcement learning^{17–19}. Recently, deep generative approaches have also been applied to metabolite annotation^{15,20,21}. Encoder-decoder neural networks were used to generate metabolite structures either from in silico predicted fingerprints (e.g., MSNovelist¹⁵) or directly from MS2 spectra (e.g., Spec2Mol²⁰ and DiffMS²¹). Besides, DeepMet²² predicted the existence of as-of-yet undiscovered metabolites through a deep generative model. Unlike in silico prediction, deep generative models do not rely on structural databases as prior knowledge, making it possible to discover metabolites beyond known chemical space. However, training such models requires large, high-quality MS2 spectral datasets. It has been estimated that even by combining all available reference spectral databases, the total would amount to only ~60,000 unique molecules¹⁵, one to two orders of magnitude lower than the ~1–2 million structures typically required to train deep generative models effectively^{23,24}. As a result, current generative models often underperform compared to in silico prediction tools in metabolite annotation, largely due to limited training data and challenges in model generalization across diverse chemical structures²¹.

Molecular networking approaches have demonstrated that in-domain knowledge, such as fragment-substructure links from neighboring known metabolites, can facilitate the discovery of unknowns²⁵. Inspired by this, we hypothesized that incorporating structure-informed knowledge into deep generation could improve de novo annotation and overcome the generalization issues of models trained on limited spectral data. In this work, we developed MetGenX, a structure-informed encoder-decoder neural network that enables efficient and controllable metabolite generation from MS2 spectra. Instead of directly encoding a query MS2 spectrum, MetGenX adopts a representation transformation strategy that first maps the query spectrum to structurally similar metabolites through spectral similarity search, and then uses these retrieved structures as templates to guide de novo generation. This approach transforms the problem from “spectrum-to-structure” to “structure-to-structure” generation, utilizing structure templates to bridge the gap between experimental data and de novo structure generation, thereby improving model performance. In independent tests, MetGenX achieves a top-1 accuracy of 55.9% on 1388 MS2 spectra from the NIST library and 68.5% on 1681 MS2 spectra from real biological samples, outperforming other in silico annotation tools in both accuracy and coverage. Notably, we applied MetGenX to untargeted metabolomics data from mouse liver tissue and identified two previously uncharacterized metabolites that are absent from human metabolome databases. Together, MetGenX establishes a flexible, structure-informed neural network that advances de novo metabolite annotation with template guidance in untargeted metabolomics, enabling the discovery of previously unknown metabolites and offering a transformative approach to biological insight.

Results

The framework of MetGenX

MetGenX is a structure-informed encoder-decoder neural network that enables efficient and controllable metabolite generation directly

from MS2 spectra. It introduces a representation transformation strategy that, instead of directly encoding the spectrum for generation, first retrieves structurally similar metabolites via spectral similarity search. These analogs serve as structure-informed templates to guide the generative model toward chemically plausible de novo structures, effectively integrating prior knowledge from known metabolites. The framework of MetGenX consists of three sequential modules: template search, template encoding, and structure generation (Fig. 1). In template search, the query MS2 spectrum is first encoded into a 512-dimensional spectral vector through textualization, word embedding and pooling (Fig. 1a and Supplementary Fig. 1a). We compared and optimized several methods for spectral embedding and template search. A Word2Vec model²⁶ was ultimately selected for its rapid embedding speed and efficient template search, while maintaining comparable structural similarity between the target structures and retrieved templates (Supplementary Fig. 1b–e). This spectral vector is then used to retrieve structurally similar compounds from a reference spectral database via cosine similarity-based match (Supplementary Fig. 1f, g). These retrieved structures serve as templates to guide de novo generation. To enable this, each structure template is encoded into a 3898-dimensional feature vector, consisting of 3878 bits for structural fingerprints, 10 bits for molecular formula, and 10 bits for formula differences (Fig. 1b and Supplementary Fig. 2). The generated feature vectors from all retrieved templates are concatenated and fed into an encoder-decoder generative model for de novo structure generation. The encoder infers the concatenated feature vector as a latent structural representation. Subsequently, this latent representation is decoded by a 6-layer transformer-based decoder to generate the final molecular structure in the form of a SMILES (Simplified Molecular Input Line Entry System) string (Fig. 1c and Supplementary Fig. 3a).

MetGenX supports both database-restricted and database-free generation modes for metabolite annotation (Fig. 1c). In the database-restricted mode, the model generates structures from a user-defined database (referred to as the background database in MetGenX). Potential candidates are first retrieved by matching the molecular formula of the target spectra to structures in the database. These candidates are then used to guide the generative model. Thus, the model generates structures based on a user-defined structure database rather than through an auto-regressive process. This mode offers high accuracy when prior knowledge of the target chemical space is available. In contrast, the database-free mode autoregressively generates structures conditioned on a specified molecular formula (Supplementary Fig. 3b). This open-ended approach enables the discovery of metabolites not present in existing databases. To further enhance annotation accuracy, a machine learning-based model re-ranks the generated candidates based on their structural similarity to the retrieved templates (Fig. 1c). Altogether, MetGenX establishes a flexible, structure-informed deep generation framework that enables de novo metabolite annotation directly from MS2 spectra.

The model training of MetGenX

MetGenX incorporates a representation transformation strategy that converts the generation task from “spectrum-to-structure” to “structure-to-structure.” Accordingly, a two-step training process is employed to achieve both “structure-to-structure” and “spectrum-to-structure” generation, respectively (Fig. 2a, b). In the first step, a transformer-based encoder-decoder neural network is pretrained on a structure database to perform a “structure-to-structure” generation task (Fig. 2a). Since we focused on the metabolite annotation in untargeted metabolomics data, the model was pretrained in biologically relevant structures. The pretraining structure database comprises 2.17 million compounds, including 0.52 million metabolites and natural products curated from widely used resources such as COCONUT²⁷, HMDB²⁸, NIST20²⁹, and GNPS⁹, as reported by Khan

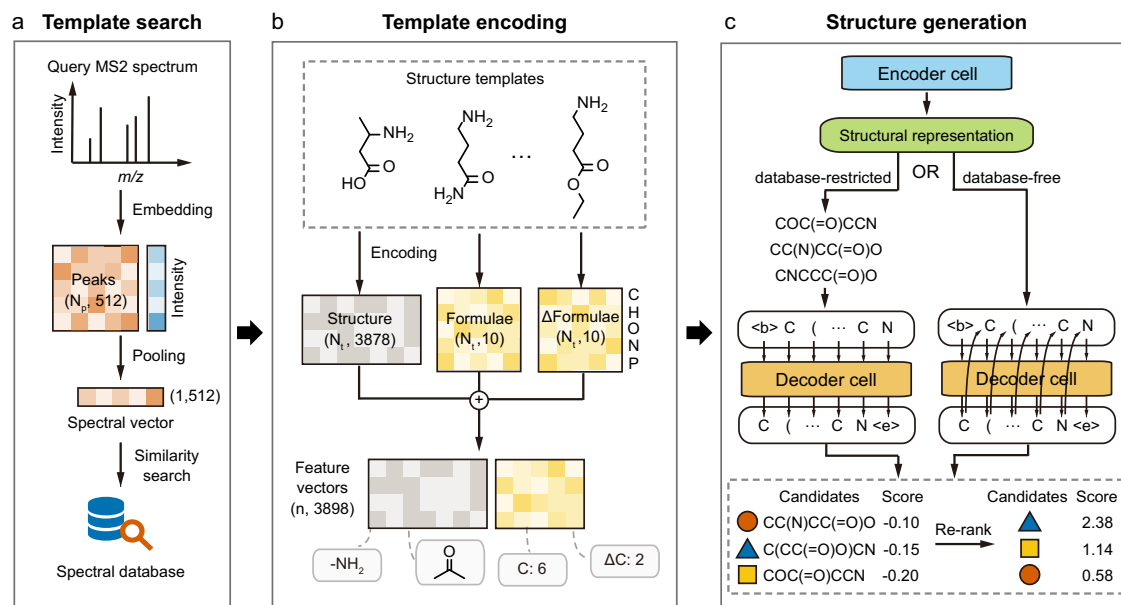


Fig. 1 | The framework of MetGenX. **a** Template search: structural templates are retrieved by performing a spectral similarity search against the query MS2 spectrum. **b** Template encoding: each structure template is encoded into a 3898-dimensional feature vector, consisting of 3878 bits for structural fingerprints, 10 bits for molecular formula, and 10 bits for formula differences. Formula differences were calculated by atom differences between the templates and the query

spectrum and also encoded as 10-bit vectors. **c** Structure generation: a structure-informed encoder-decoder neural network that supports both database-restricted and database-free generation modes for metabolite annotation. A machine learning-based model re-ranks the generated candidates based on their structural similarity to the retrieved templates.

et al.³⁰, along with an additional 1.76 million biologically relevant structures retrieved from PubChem³¹. To enable structure-informed generation, each query structure is first matched against the background database (including 0.52 million metabolites and natural products) to retrieve structurally similar templates based on structural similarity. These templates are then encoded and used to guide de novo molecular generation. The pretraining process converged after 33 epochs training, with early stopping triggered if the validation loss did not improve for 5 consecutive epochs (see “Methods”). In the second step, the pretrained model is further fine-tuned for the “spectrum-to-structure” generation task using a reference spectral database (Fig. 2b). The fine-tuning is not intended to replicate pretraining, but rather to bridge the gap between structure and spectra, ensuring the model is well aligned and optimized for metabolite annotation from spectra. In this study, the NIST spectral database comprising 23,213 compounds was utilized for fine-tuning. The metrics between the two datasets indicated the structural consistency between the pretraining and fine-tuning datasets, although some chemical subspaces remain uncovered (Supplementary Fig. 4). The NIST spectral database was partitioned into three subsets: a training set (19,541 spectra), a validation set (2172 spectra), and a testing set (1500 spectra). As outlined in the MetGenX framework (Fig. 1), each query spectrum was first encoded and used to retrieve structure templates in a reference spectral database via spectral similarity search. These retrieved structures were then employed as templates to guide the de novo generation. The fine-tuning process triggered with early stopping if no improvement in validation loss was observed for 5 consecutive epochs, and the model at epoch 4 was selected as the best model. To evaluate the contribution of each model component, we conducted a series of ablation experiments focusing on structure pretraining, template search, and formula constraint (Supplementary Fig. 5a–c). We assessed their impact on generation accuracy under both database-restricted and database-free settings. The results highlighted the critical roles of structure pretraining and template search, both of which significantly enhance the accuracy of generation.

To further enhance annotation accuracy, we developed a machine learning-based re-ranking model that prioritizes generated candidates according to their structural similarity to retrieved templates (Fig. 2c). The model is trained on generated candidate-template pairs, with twelve characterization features calculated for each pair and used as input features for a LightGBM model. These features include the relationship between the generated structures and their templates (distributions of spectral similarity scores and structure similarity scores, and number of templates), model score, candidate rank and the molecular descriptors for generated structures (molecular complexity, NP score and SMILES length), which were empirically selected to improve the ranking performance and ensure the reliability of the generated results. Since the re-ranking model depends on the templates, the ability of MetGenX to fully explore previously uncharacterized chemical scaffolds may be potentially limited. Feature importance analysis revealed that scores related to structural similarity contribute most significantly to model performance, underscoring the critical role of template-based structural information in candidate ranking (Supplementary Fig. 5d, e). Altogether, the pretraining-finetuning framework combined with re-ranking enables MetGenX to achieve accurate and robust metabolite annotation, while also empowering the discovery of previously uncharacterized metabolites.

The performance evaluation

An independent testing set of 1500 spectra from the NIST spectral database was used to evaluate the performance of MetGenX (Fig. 3a). None of these compounds were included in the model’s training or optimization. As shown in Fig. 3b, the majority of spectra (92.5%) successfully retrieved at least one structure template from the reference spectral database (the NIST training set). For each query spectrum ($n=1388$), candidate structures were firstly generated in the database-restricted mode and subsequently re-ranked using MetGenX. The background database comprised 0.52 million chemical structures. As shown in Fig. 3c, MetGenX achieved a top-1 accuracy of 48.9% and a

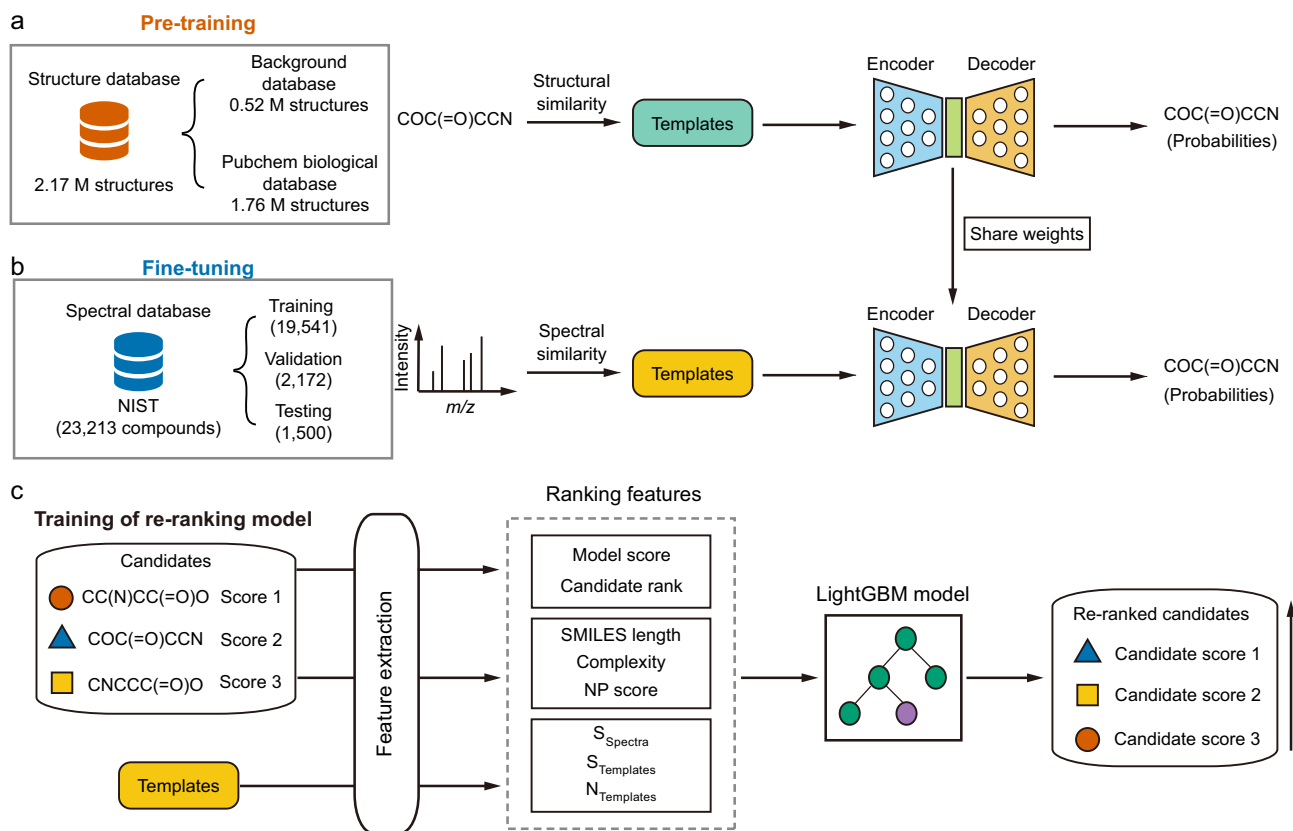


Fig. 2 | The model training of MetGenX. a, b Pretraining–finetuning framework: the generative model is first pretrained on a structure database with a total of 2.17 million structures (a), and then fine-tuned by a spectral database from NIST (b).

c Generated candidates are re-ranked using a LightGBM model based on their structural similarity to retrieved templates. The model outputs candidate scores (S) after ranking.

top-3 accuracy of 71.8% prior to re-ranking. After re-ranking, performance improved notably, with top-1 and top-3 accuracies increasing to 55.9% and 76.1%, respectively (Supplementary Data 1). Positional performance was further evaluated using the relative rank position (RRP)³², with the density of RRP biased toward 1.0, indicating high ranking accuracy (Fig. 3d). These results indicate that MetGenX is capable of generating correct structures when prior knowledge of the target chemical space is available.

To further assess MetGenX's ability to generate structures without constraint of a reference chemical space, we evaluated its performance in a database-free generation mode. As shown in Fig. 3e, MetGenX achieved a top-1 accuracy of 21.7% and a top-3 accuracy of 35.0% in the NIST testing set (Supplementary Data 2). Although the information contained in the query spectrum may be insufficient for the de novo generation of the exact structure, MetGenX is capable of generating structurally similar candidates. Among the top-3 candidates, 75.0% exhibited high structural similarity to the target structures, indicating that the generated matches are meaningful and can offer valuable insights for further discovery (Fig. 3f). We assessed the robustness of model training by retraining the generative model with five random seeds on the NIST dataset (Supplementary Fig. 6a, b). The consistent results and low variance confirm the stability of the training process.

As a structure-informed generative model, MetGenX captures domain knowledge from structure templates and generates corresponding structures. This principle is illustrated by examples shown in Fig. 3g, h. For example, given the query spectrum of pyridoxamine, pyridoxine derivatives, such as pyridoxine and 4-methoxypyridoxine, were retrieved as templates from the reference spectral database. Using these templates as input, MetGenX successfully generated pyridoxamine as the top-1 candidate in the database-restricted mode (Fig. 3g). Another example of 2-ethoxyaniline was generated in the

database-free mode (Fig. 3h). Similarly, the template search retrieved four structural templates, enabling the generation of 2-ethoxyaniline as the top-1 candidate.

Additionally, we systematically investigated the relationship between structure templates and model performance. In both database-restricted and database-free generation modes, a larger number of templates and higher structural similarity between the templates and target structures were associated with the improved accuracy of structure generation (Supplementary Fig. 6c–k). Across all retrieved templates, the mean maximum similarity was 0.65, indicating that MetGenX could effectively retrieve at least one structurally relevant template for most spectra. The mean similarity across all retrieved templates was 0.43, reflecting moderate overall similarity across all retrieved templates (Supplementary Fig. 6f). Specifically, we then evaluated different thresholds of maximum spectral and structural similarity (Supplementary Fig. 6g, h). Maximum spectral similarity showed no clear correlation with prediction correctness, whereas maximum structural similarity exhibited strong discriminative power. For example, when the maximum structural similarity exceeded 0.71 (determined by Youden's J index), the probability of correct generation markedly increased (true positive rate 0.60 and false positive rate 0.33). These results indicated that a high maximum structural similarity is needed for high-accurate structure generation. Overall, these results demonstrate that MetGenX achieves effective structure generation for metabolite annotation and discovery through a structure-informed deep generation framework.

Annotation of metabolites in real biological samples

We further evaluated the performance of MetGenX in annotating metabolites using experimental spectra acquired from real biological samples (Fig. 4a). LC-MS-based untargeted metabolomics data were

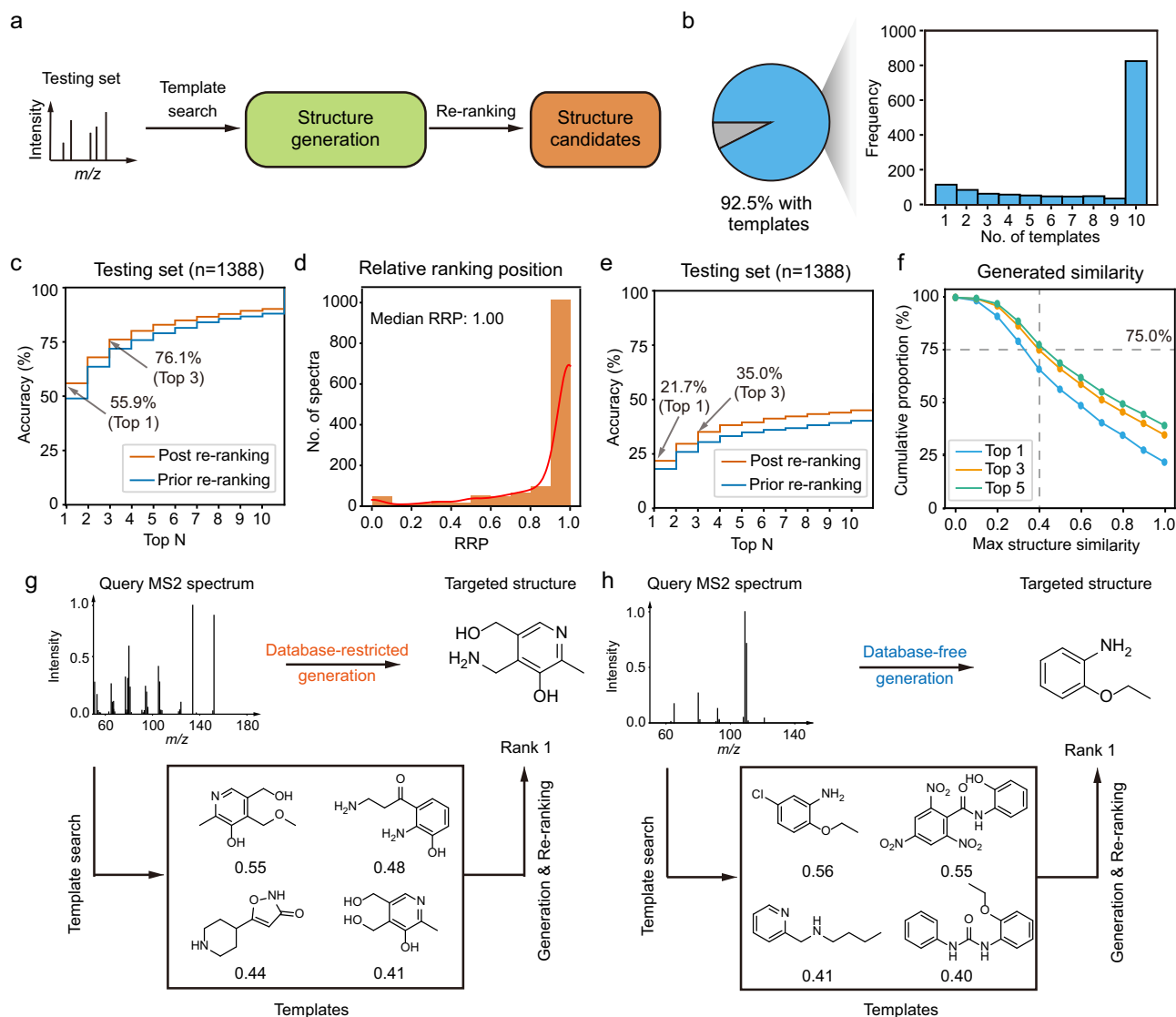


Fig. 3 | The performance evaluation of MetGenX. a An independent set of 1500 spectra from the NIST spectral database was used to evaluate the performance of MetGenX. **b** Distribution of template numbers retrieved from the reference spectral database. **c** Annotation accuracies of the database-restricted generation before and after re-ranking using MetGenX. **d** Distribution of relative ranking position (RRP). The orange line represents the density distribution of RRP. **e** Annotation accuracies of the database-free generation before and after re-ranking

using MetGenX. **f** Cumulative percentages of maximum structural similarity between generated structures (top-1, top-3 or top-5) and target structures. **g, h** Two generation examples: pyridoxamine (**g**) through the database-restricted mode, and 2-ethoxyaniline (**h**) through the database-free mode. Numbers below each template indicate the spectral similarity between the query MS2 spectrum and the corresponding templates. Source data are provided as a Source Data file for (**b–f**).

collected from five sample types: cells, brain tissue, liver tissue, plasma, and urine. The metabolite annotation in biological samples was performed based on matching m/z , RT, and MS/MS spectral similarity against in-house standard library curated from chemical standards, representing level 1 annotation according to Metabolomics Standards Initiative³³ (Fig. 4b and Supplementary Data 3). Across these samples, we identified 1720 spectra in total. MetGenX was then applied to annotate these spectra. Here, the full NIST spectral dataset was used as the reference spectral database. Among them, 1681 spectra (97.7%) successfully retrieved at least one structure template from the reference spectral database and were subsequently used for structure generation (Fig. 4b and Supplementary Fig. 7a, b). Because most metabolites in biological samples have been recorded in a structure database, we employed the database-restricted generation mode for annotation. The background database comprised 0.52 million chemical structures. For these spectra, MetGenX achieved a top-1 accuracy of 68.5% and a top-3 accuracy of 89.2% (Fig. 4c and Supplementary Fig. 7c,

d). Furthermore, we calculated the annotation accuracy counted by metabolites in Supplementary Fig. 7e. For the mean rank of these metabolites, MetGenX achieved a top-1 accuracy of 64.9% and a top-3 accuracy of 85.3%. Among the three major adducts, top-3 accuracies reached 90.5% for $[M+H]^+$, 76.8% for $[M-H_2O+H]^+$, and 60.0% for $[M+Na]^+$ (Fig. 4d). Additionally, consistent annotation accuracy was observed across a wide range of chemical subclasses. Notably, MetGenX performed especially well for lipids, likely due to their structural regularity and the high availability of related templates (Fig. 4e). Furthermore, since the collision energy used for biological samples differed from the merged collision energies in the reference spectral database, we searched templates with spectra acquired at NCE30, which better matched our experimental conditions. We found that using NIST standard spectra acquired at NCE30 for template retrieval, which matched the experimental conditions of biological samples, improved generation coverage and accuracy (Supplementary Fig. 7f, g).

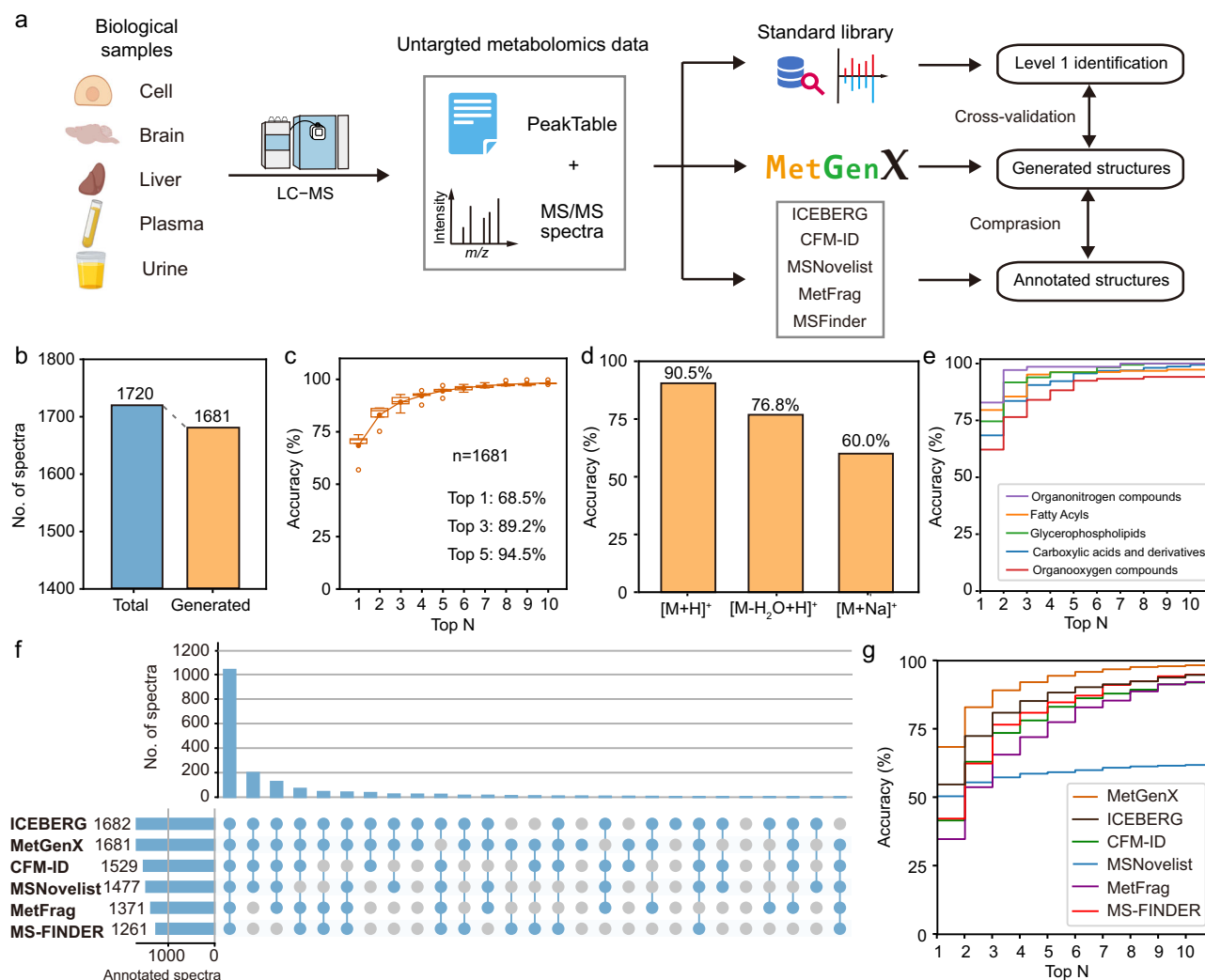


Fig. 4 | Annotation of metabolites in real biological samples. **a** Workflow for metabolite annotation in biological samples with different tools. **b** Number of annotated spectra across different samples using library-based spectral match (level 1 annotation; blue) and MetGenX (orange). **c** Top-N accuracy of MetGenX stratified by sample types. The large dot inside each box and orange line indicate the overall accuracy combining all spectra. Box plots show the median (center line), 25th and 75th percentiles (box bounds). The whiskers extend to the most extreme data points within $1.5 \times$ the interquartile range (IQR) from the lower and upper

quartiles. Data points beyond the whiskers are plotted individually as outliers. **d, e** Annotation accuracy stratified by adduct types (**d**) and chemical subclasses (**e**). **f, g** Comparison of annotation coverage (**f**) and annotation accuracy (**g**) across different in silico annotation tools. Connected filled dots indicate the tools included, and bars show the shared spectra annotated by those tools. Source data are provided as a Source Data file for (**b–g**). Illustrations in (**a**) created by BioRender. Zhu, Z. (2026) <https://BioRender.com/fgn01p9>.

We further benchmarked the performance of MetGenX against several widely used in silico annotation tools, including CFM-ID¹¹, MS-FINDER¹², MetFrag³⁴, MSNovelist¹⁵, and ICEBERG³⁵ (Fig. 4a). To ensure a fair comparison, all tools used the same background database comprising 0.52 million chemical structures, which referred during model training. The background database was collected from the public repository, such as COCONUT²⁷, HMDB²⁸, NIST20²⁹, and GNPS⁹, which were widely used in untargeted metabolomics studies. Across the combined dataset, MetGenX outperformed all other tools in annotation accuracy (Fig. 4f, g). Annotation coverage was defined as the number of MS2 spectra for which at least one valid and distinguishing candidate structure was successfully generated. Specifically, MetGenX successfully annotated 1681 out of 1720 spectra (97.7%), showing comparable annotation coverage with ICEBERG (1682, 97.8%) and significantly exceeding the performance of other tools, which annotated between 1261 (73.3%; MS-FINDER) and 1529 (88.9%; CFM-ID) spectra (Fig. 4f, Supplementary Fig. 8a). In terms of accuracy, MetGenX achieved a top-3 accuracy of 89.2%,

outperforming other tools whose top-3 accuracies ranged from 57.4% (MetFrag) to 81.0% (ICEBERG; Fig. 4g). The accuracy based on both the annotation results shared by all tools (shared annotations; Supplementary Fig. 8b) and the complete dataset (all inputted spectra; Supplementary Fig. 8c) showed consistently higher top-3 accuracy than other tools in biological samples, demonstrating its robustness across different evaluation conditions. This performance advantage stems from the structure-informed generation framework of MetGenX, which retrieves structurally similar templates to better accommodate diverse fragmentation patterns. Compared to deep generative models like MSNovelist, a background database can narrow down the search space and enhance accuracy. Unlike prediction methods such as CFM-ID, ICEBERG, MS-FINDER, and MetFrag, MetGenX avoids the limitations of unmatched or unexplained fragment peaks, enabling both high coverage and accuracy. In addition, we randomly selected ten metabolite examples annotated by MetGenX that were absent from the NIST spectral database to demonstrate successful structure generation (Supplementary Fig. 9). Altogether,

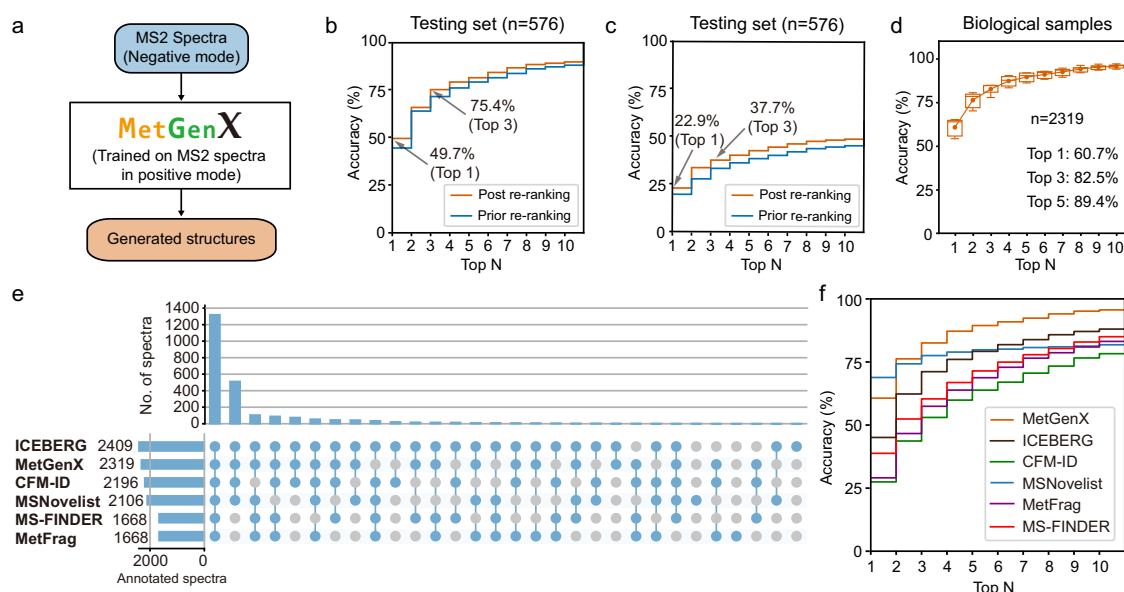


Fig. 5 | Metabolite generation in negative ionization mode. **a** Workflow for metabolite generation through MetGenX in negative mode without additional retraining. **b, c** Annotation accuracies of the database-restricted generation (**b**) and database-free generation (**c**) before and after re-ranking using MetGenX. **d** Top-N accuracy of MetGenX stratified by sample types. The large dot inside each box and orange line indicate the overall accuracy combining all spectra. Box plots show the median (center line), 25th and 75th percentiles (box bounds). The whiskers extend

to the most extreme data points within $1.5 \times$ the interquartile range (IQR) from the lower and upper quartiles. Data points beyond the whiskers are plotted individually as outliers. **e, f** Comparison of annotation coverage (**e**) and annotation accuracy (**f**) across different in silico annotation tools. Connected filled dots indicate the tools included, and bars show the shared spectra annotated by those tools. Source data are provided as a Source Data file for (**b–f**).

these results demonstrate the strong capability of MetGenX to annotate metabolites lacking reference tandem mass spectra.

Metabolite generation in negative ionization mode

Metabolite annotation in negative ionization mode is crucial yet remains uncharacterized in untargeted metabolomics. Most in silico tools face challenges in this mode due to the limited availability of reference spectra and insufficient training data. MetGenX addresses this limitation through its “structure-to-structure” generative framework, which leverages retrieved structural templates to guide de novo generation. This design allows the generative model to be applied directly to different ionization modes without additional retraining (Fig. 5a). To evaluate the generalization capability of MetGenX in negative ionization mode, we first tested it on an independent set from the NIST spectral database ($n = 653$ spectra). Among them, 576 spectra (88.2%) had successfully retrieved at least one structure template from the reference spectral database. In database-restricted generation mode, MetGenX achieved a top-1 accuracy of 49.7% and a top-3 accuracy of 75.4% after re-ranking (Fig. 5b and Supplementary Data 4). In database-free generation mode, the top-1 and top-3 accuracies were 22.9% and 37.7%, respectively (Fig. 5c and Supplementary Data 5). The performance metrics closely matched those observed in positive ionization mode, further demonstrating MetGenX’s robust generalization across different ionization modes.

We further assessed MetGenX and other in silico tools using experimental data from real biological samples (Fig. 5d–f, Supplementary Data 6). MetGenX successfully annotated 2319 out of 2417 spectra (95.9%), achieving a top-1 accuracy of 60.7% and a top-3 accuracy of 82.5% (Fig. 5d). The annotation coverage of other tools ranged from 1668 (69.0%; MetFrag) to 2409 (99.7%; ICEBERG) spectra (Fig. 5e, Supplementary Fig. 8d). For top-3 accuracy, MetGenX significantly outperformed CFM-ID (53.0%), MetFrag (57.4%), MS-FINDER (60.4%) and ICEBERG (71.2%). The performance gap between positive

and negative ionization modes observed in existing tools underscores their limited generalizability. Compared to MSNovelist, MetGenX achieved higher accuracy beyond the top-1 rank, reaching a top-3 accuracy of 82.5% versus 77.5%. Similar to positive ionization mode, the accuracy based on both the annotation results shared by all tools (shared annotations; Supplementary Fig. 8e) and the complete dataset (all inputted spectra; Supplementary Fig. 8f) showed consistently higher top-3 accuracy than other tools in biological samples, demonstrating its robustness across different evaluation conditions. Collectively, these results demonstrate that MetGenX’s structure-informed generative framework enables robust performance across ionization modes without the need for retraining.

To further verify the annotation performance in real biological samples, we spiked 200 metabolite standards into mouse liver and NIST 1950 plasma samples. Among the 177 spiked metabolites detected and identified in mouse liver tissue, MetGenX achieved a top-1 accuracy of 76.8% and a top-3 accuracy of 92.1%, outperforming other tools, whose top-3 accuracies ranged from 59.9% to 80.2% (Supplementary Fig. 7h). Similarly, for the 166 spiked metabolites detected and identified in NIST plasma samples, MetGenX achieved a top-1 accuracy of 74.1% and a top-3 accuracy of 89.2%, exceeding the performance of other tools, whose top-3 accuracies ranged from 60.2% to 75.9% (Supplementary Fig. 7i). This spike-in validation results closely aligned with those obtained from endogenous metabolites in biological samples (top-3 accuracy of 89.2% and 82.5%, Figs. 4c and 5d), further confirming the robustness and reliability of MetGenX annotations.

To further compare MetGenX with network-based methods, we evaluated its performance against MetDNA3 across 20 diverse biological sample datasets. For MetDNA3, 30% of level 1 annotations were used as seeds for network propagation, with the remaining 70% reserved for validation under a 10-fold cross-validation, as previously reported³⁶. For MetGenX, all annotated structures were excluded during template search, and the same metabolic reaction network

used by MetDNA3 was adopted as the background database for database-restricted generation. Across these datasets, MetGenX achieved higher annotation coverage (84.2% vs. 68.1%) but slightly lower accuracy (78.1% vs. 84.4%) compared with MetDNA3 (Supplementary Fig. 7j, k). Considering all spectra, MetGenX achieved a higher overall identification rate than MetDNA3 (65.9% vs. 57.6%, Supplementary Fig. 7l). These results indicate that MetGenX's database-restricted mode provides broader annotation coverage, whereas MetDNA3 is better suited for high-confidence annotation. This difference is expected, as MetDNA3 integrates multiple sources of contextual information (e.g., predicted retention time), while MetGenX relies solely on MS2 spectra to rank candidates. Overall, these results substantiate the practical value of MetGenX's database-restricted mode and clarify its complementary role relative to MetDNA in metabolite annotation.

Metabolite discovery in mouse liver tissue

To demonstrate MetGenX's applicability to both known and unknown metabolites, we analyzed untargeted metabolomics data from mouse liver. In positive ionization mode, 6564 metabolic features were acquired with MS2 spectra, and 2979 of them were confidently assigned molecular formulae using SIRIUS³⁷ (Fig. 6a, Supplementary Data 11). A multi-step annotation workflow was applied to these spectra, proceeding sequentially through library-based spectral matching, database-restricted generation, and database-free generation using MetGenX. Among the 2979 spectra, 516 were annotated through library-based spectral matching and classified as level 1 annotations. The remaining 2463 spectra were subjected to database-restricted generation with MetGenX. Among them, 2161 spectra (87.74%) successfully retrieved at least one structural template. The database-restricted mode generated high-quality candidate structures for 541 spectra (candidate score > 1.2). The remaining spectra were annotated using database-free generation, resulting in 426 spectra with high-quality candidates and 1102 spectra with at least one valid structure. A total of 394 spectra failed to yield valid structures, likely due to no suitable templates retrieved or incorrect formula assignments. Notably, the natural product scores³⁸ of high-quality candidates from both generation modes closely resembled those of standard annotations, indicating that MetGenX effectively generates metabolite-like structures (Supplementary Fig. 10c).

We further validated the unknown metabolites annotated by MetGenX in mouse liver. One feature (M371T394; m/z = 371.1131 Da; RT = 394 s) was assigned the molecular formula $C_{13}H_{18}N_6O_5S$ and annotated as adenosyl-cysteine through database-restricted generation (Fig. 6b). Notably, adenosyl-cysteine is a metabolite absent from major metabolome databases such as HMDB and KEGG. This candidate was the top-1 ranked generated structure (candidate score = 2.0), generated using a set of nucleotide-related templates (Supplementary Fig. 10d). The major fragments in the MS2 spectrum were manually interpreted, and the annotation was further confirmed using a synthesized chemical standard (Fig. 6c and Supplementary Fig. 10e). This HMDB-absent metabolite was detected in different mouse tissues, with the highest enrichment observed in the kidney and liver tissue (Fig. 6d).

Another feature (M678T439; m/z = 678.1066 Da; RT = 439 s) was assigned the molecular formula $C_{20}H_{25}N_9O_{14}P_2$, which is not present in the background database. Consequently, this feature was annotated using database-free generation. Template search retrieved a set of nucleotide-related metabolites, suggesting that the unknown compound is a nucleotide derivative (Supplementary Fig. 10f). MetGenX subsequently generated adenosine 5'-diphosphate 5'-inosine (Ap2I) as one of the top-3 candidate structures (candidate score = 2.6; Fig. 6e), which was prioritized for synthesis and validation based on its biochemical plausibility. The major fragments in the MS2 spectrum were manually interpreted, and the annotation was further confirmed using

a synthesized chemical standard (Fig. 6f, Supplementary Figs. 10g and 11a). Ap2I is a previously uncharacterized dinucleoside polyphosphate compound that is absent from PubChem. This metabolite was detected in multiple mouse tissues, with the highest enrichment observed in the spleen and liver (Fig. 6g). To verify that Ap2I represents a real metabolite in biological samples, several validation experiments were performed. The synthetic Ap2I standard exhibited identical retention time and highly similar MS2 spectra to the biological peak in both ionization modes (Fig. 6f and Supplementary Fig. 11a). Mixing nucleotide precursors (AMP and IMP) did not yield any Ap2I signal in either positive and negative ionization mode (Supplementary Fig. 11b), indicating that Ap2I was not formed in the ion source. Moreover, Ap2I showed distinct retention times from potential dimer components (AMP, IMP, ADP, and ATP) in both ionization modes (Supplementary Fig. 11c). Furthermore, fragmentation-energy ramping experiments showed that the precursor intensity of Ap2I decreased with increasing collision energy, accompanied by the appearance of fragment ions, consistent with the fragmentation patterns observed using the chemical standard (Supplementary Fig. 12). Collectively, these results demonstrate that Ap2I is a true metabolite present in biological sample.

Additionally, we quantified the concentration of two discovered metabolites in mouse liver tissue using the synthetic standards (Supplementary Fig. 13). As a result, the mean concentrations were 10.80 ± 0.10 pmol/mg tissue for adenosyl-cysteine and 2.97 ± 0.92 pmol/mg tissue for Ap2I. Together, these findings highlighted the powerful capacity of MetGenX to uncover previously uncharacterized chemical entities and expand the metabolome.

We further evaluated the annotation performances using a more constrained or biologically relevant structural database³⁶ containing 66,874 metabolites from NIST20, KEGG, HMDB and MetaCyc. Using this database, 429 spectra were annotated in the database-restricted mode, and 1640 spectra were annotated in the database-free mode. The overall annotation coverage exhibited a similar distribution between the two databases (Supplementary Fig. 14). To assess the impact of background databases, we compared the database-restricted results annotated through a larger background database (541 spectra in total). After switching to the biologically relevant database, 81.9% of the annotations remained consistent, while 18.1% had different annotation results across the top-3 annotations. This finding highlights the importance of incorporating prior knowledge and using a specific compound database to further improve annotation accuracy and reduce false discovery rates.

Benchmark evaluation of MetGenX in public dataset

To further validate the generalization and scalability of MetGenX, we benchmarked MetGenX on the public spectral dataset MassSpecGym³⁹. Since MassSpecGym defines distinct tasks corresponding to different application scenarios, we retrained MetGenX following the standard MassSpecGym workflow and evaluated performance under both database-restricted and database-free modes with corresponding baseline models. In database-restricted mode, molecular retrieval metrics in MassSpecGym were applied and MetGenX generated structures based on the MassSpecGym reference structure database. To ensure a fair and meaningful comparison, we aligned MetGenX with this retrieval task. Specifically, for each query spectrum, MetGenX was constrained to generate and rank structures only from the same candidate lists provided by MassSpecGym. This restriction ensures that MetGenX and all baseline methods operate on an identical search space, effectively converting the database-restricted generation into a molecular retrieval task. On the MassSpecGym testing set, MetGenX achieved a top-1 accuracy of 18.34%, top-5 accuracy of 31.39% and top-20 accuracy of 48.43% (Supplementary Table 1). Compared with baseline models reported by MassSpecGym, the top-1 accuracy of MetGenX (18.34%) outperformed MIST

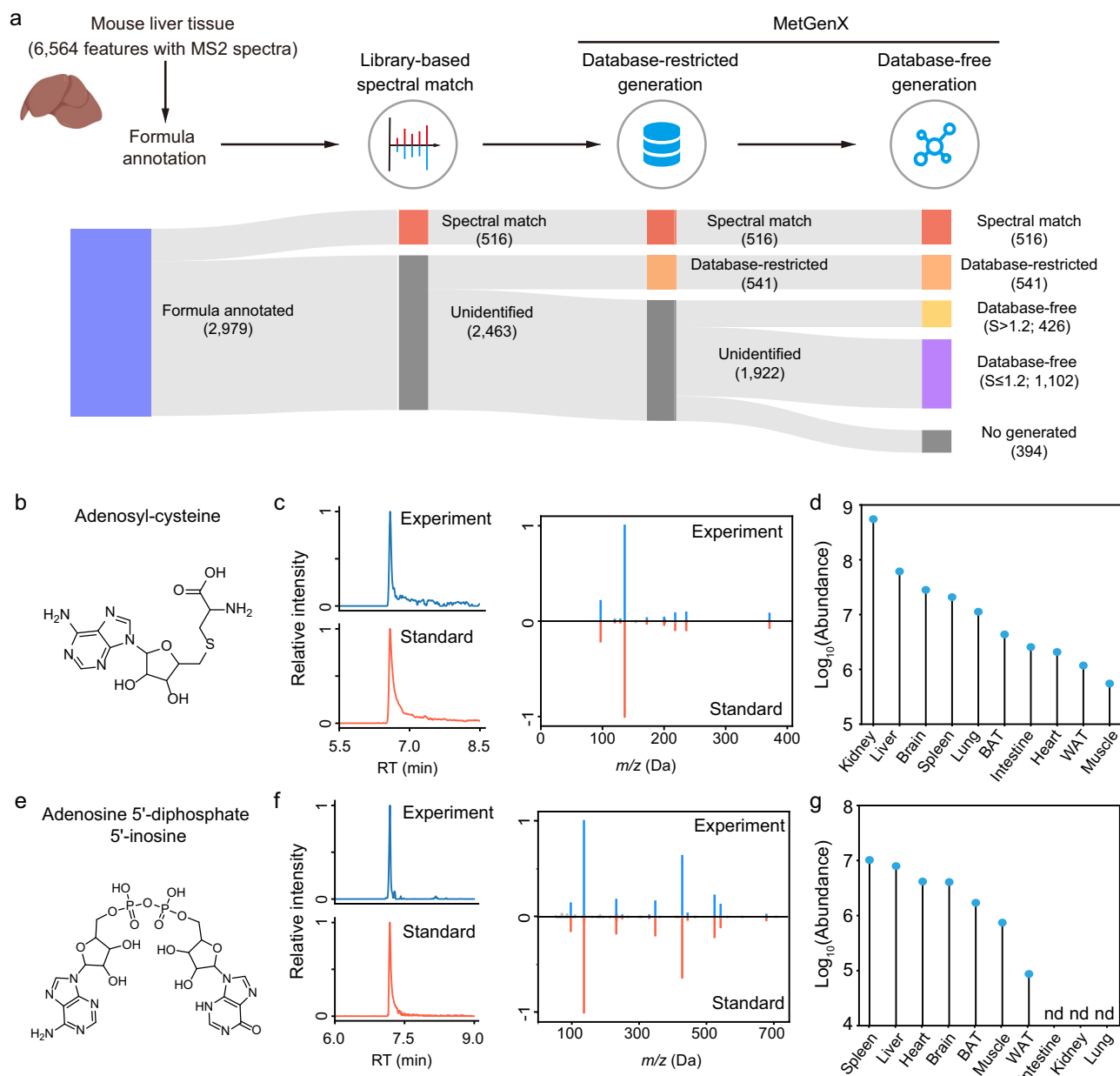


Fig. 6 | Metabolites discovery in mouse liver tissue. **a** A multi-step annotation workflow was applied to annotate untargeted metabolomics data of mouse liver tissue. S indicates the candidate score of MetGenX. **b** Structure of adenosyl-cysteine. **c** Standard validation of adenosyl-cysteine using a chemical standard by retention time and MS2 spectral matches. **d** Abundance distributions of adenosyl-cysteine in different tissues of mice. BAT brown adipose tissue, WAT white adipose

tissue. **e** Structure of adenosine 5'-diphosphate 5'-inosine (Ap2I). **f** Standard validation of Ap2I using a chemical standard by retention time and MS2 spectral matches. **g** Abundance distributions of Ap2I in different tissues of mice. nd not detected. Source data are provided as a Source Data file for (a, d and g). Illustrations in (a) created by BioRender. Zhu, Z. (2026) <https://BioRender.com/fgn01p9>.

(9.57%), DeepSets + Fourier features (6.56%), and Fingerprint FFN (5.09%), demonstrating its high performance when annotating structures within a defined chemical space. Alternatively, in database-free mode, MetGenX generated structures without constraint of a reference structure database and compared with baseline models reported by literatures^{21,39}. On the MassSpecGym testing set, MetGenX achieved a top-1 accuracy of 2.50% and top-10 accuracy of 4.62% (Supplementary Table 2). MetGenX outperformed other generative models, including MSNovelist, Spec2Mol and Neuraldecipher. Notably, these generative models failed to generate any correct structures, with zero top-1 accuracy on the MassSpecGym testing set, which highlights the inherent difficulty of structure generation in the MassSpecGym dataset³⁹. Moreover, the top-10 accuracy of MetGenX (4.62%) was

higher than MADGEN⁴⁰ (1.54%). Compared with DiffMS²¹, MetGenX showed comparable performance with higher top-10 accuracy (4.62% vs. 4.25%).

Additionally, we performed an additional experiment on the MassSpecGym dataset by removing the similarity score threshold of 0.4 in the template search of MetGenX and instead selected the ten structure templates with the highest similarity scores as inputs for de novo structure generation of MetGenX. With this revision, all input MS2 spectra could be annotated by MetGenX. This configuration is denoted as MetGenX-Full. Although this leads to a modest performance decrease, MetGenX-Full still achieves accuracy comparable to or outperforms other baseline models (Supplementary Tables 1 and 2). Together, this demonstrates that MetGenX's performance is not driven

by selective coverage. The MetGenX-Full model is provided only for the MassSpecGym benchmark to achieve complete annotation coverage across all spectra. In practical applications, the template similarity cutoff remains necessary to ensure reliable structural generation.

It is expected that MetGenX exhibits relatively lower absolute accuracies on the MassSpecGym dataset. This is because the dataset enforces that no test molecule shares a maximum common edge substructure (MCES) smaller than 10 with any training molecule, making template-based structure generation particularly challenging. Nevertheless, MetGenX maintained competitive performance across all evaluation metrics compared with other annotation tools, demonstrating its robustness and generalizability across structurally diverse datasets.

Discussion

Metabolite annotation, especially for unknown metabolites, remains a fundamental challenge in untargeted metabolomics due to the limited availability of reference MS2 spectra and the vast structural diversity of the metabolome. To address this, we developed MetGenX, a structure-informed encoder-decoder neural network that generates candidate metabolite structures guided by structurally similar templates. Inspired by molecular networking, MetGenX reformulates the conventional spectrum-to-structure task into a structure-to-structure generation problem using a representation transformation strategy. This enables the model to be trained on over two million chemical structures, greatly exceeding the scale of available reference MS2 spectra. As a result, it effectively bridges the gap between sparse spectral data and the demands of a deep generative model. Structural templates enhance the fidelity of structure generation and enable a re-ranking model to prioritize chemically plausible candidates. Unlike existing *in silico* tools that rely on direct interpretation of fragment peaks^{12,34}, MetGenX infers structural information based on structural similarity to known templates. By leveraging prior structural knowledge embedded in these templates, MetGenX achieves higher annotation accuracy and broader metabolome coverage. MetGenX was designed to generate chemical structures from structural templates. Thus, the structural coverage of the spectral libraries used for template retrieval may limit its ability to produce metabolites with chemical structures that are entirely outside known chemical space. Consequently, a small fraction of spectra (<10%; Fig. 3b and Supplementary Fig. 7b) cannot be processed without suitable templates in the spectral libraries. As spectral libraries such as NIST continue to expand and incorporate more chemical standards, these limitations are expected to diminish, leading to improved generation coverage and accuracy.

A major advantage of MetGenX is its flexibility to support both database-restricted and database-free generations. In the database-restricted generation, the model generates candidate structures from a known structure database (0.52 million compounds), which narrows the search space and improves annotation accuracy. In contrast, the database-free generation enables MetGenX to generate compounds not found in any known chemical database, supporting the discovery of previously uncharacterized metabolites. In this mode, the templates provide structural guidance rather than directly selected as generated structures. This design preserves the *de novo* nature during structure generation. Consequently, the performance of the database-free mode is expectedly lower than that of the database-restricted mode. In both modes, the retrieved templates serve only as guiding signals to inform the generative process rather than as direct structural sources. This unique design addresses limitations of traditional generative models and enhances the ability to annotate unknown regions of the metabolome. This method is suitable for tasks with or without a targeted chemical space, and can be complementary in metabolite annotation in biological samples. To demonstrate this utility, we implemented a multi-step annotation

workflow combining the library-based spectral matching, database-restricted generation, and database-free generation using MetGenX to analyze real untargeted metabolomics data from biological samples (Fig. 6). The database-restricted mode enables reliable identification of known metabolites, while the database-free mode explores chemical space beyond existing databases to predict structures absent from known chemical space. Notably, although MetGenX demonstrates strong annotation performance in terms of coverage and accuracy, top-1 accuracy remains the most relevant metric for practical metabolite annotation workflows. Higher-rank accuracies (e.g., top-3 or top-10) are therefore primarily intended for methodological benchmarking.

Another key strength of MetGenX is its flexibility and transferability across diverse application scenarios. Built on a structure-to-structure generation framework, MetGenX is compatible with a wide range of experimental MS2 spectra and demonstrates strong generalization capabilities. In this study, we pretrained the model on a large molecular structure dataset and fine-tuned it for spectral annotation tasks. Ablation experiments confirmed that pretraining significantly improved downstream performance (Supplementary Fig. 5), underscoring the model's ability to transfer structural knowledge to MS2 spectrum-based annotation. Notably, despite being trained exclusively on tandem MS spectra from the positive ionization mode, MetGenX generalized effectively to tandem MS spectra from the negative ionization mode without retraining. This highlights its robustness and adaptability, enabling seamless application to new or customized spectral libraries without modifying the generative model. Furthermore, for MS2 spectra from different adducts, MetGenX processes all adduct types through a unified workflow without explicitly encoding adduct types. Spectra are searched against the spectral library based on spectral similarity, and the retrieved templates guide structure generation through the structure-informed model. This design enables MetGenX to generalize across adduct types and generate valid molecular structures even for unseen adducts.

In recently years, the network-based methods, such as MetDNA³⁶ was applied in metabolite annotations through a given knowledge-based structural database. Those network-based approaches showed great complementarity of our methods. On the one hand, adding relationships between metabolites and unknown spectra in biological samples may help to improve the annotation coverage and accuracy. On the other hand, the generation model presents opportunities for enlarging knowledge databases of previously uncharacterized structures. Moving forward, combining generative models with network-based metabolite annotation will be a promising approach for the discovery of unknown metabolites.

To demonstrate its real-world utility, we applied MetGenX to untargeted metabolomics data from mouse liver tissue using the multi-step annotation workflow. This analysis led to the identification of two previously unreported metabolites (adenosyl-cysteine and adenosine 5'-diphosphate 5'-inosine) absent from major human metabolome databases. These findings validate MetGenX's capability to uncover metabolites absent in present database through structure generation. Together, MetGenX establishes a flexible, knowledge-guided generative framework for metabolite annotation and unknown metabolite discovery, pushing the boundaries of current untargeted metabolomics.

Methods

Ethical statement

The animal experiments were compliant with the ethical guidelines of the Institutional Animal Care and Use Committees of Interdisciplinary Research Center on Biology and Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences (approval research project number: ECSIOC_2023-23).

Preparation of the training data

The training of MetGenX consisted of two stages: pretraining on a structure dataset and fine-tuning on a spectral dataset (Fig. 2). The pretraining structure dataset comprises 2.17 million compounds, including 0.52 million metabolites and natural products curated from widely used resources such as COCONUT²⁷, HMDB²⁸, NIST20²⁹, and GNPS⁹, as reported by Khan et al.³⁰, along with an additional 1.76 million biologically relevant structures retrieved from PubChem³¹. All structures were deduplicated based on the first layer of the InChI-Key, and their SMILES were standardized by removing stereochemical information to obtain canonical SMILES.

For fine-tuning, MS2 spectra acquired in positive ionization mode using Orbitrap instruments were collected from the NIST20 spectral database. MS2 spectra lacking InChIKey or mismatching between molecular formula and SMILES were excluded. The dataset was further curated using the following criteria: precursor m/z between 50 and 1200, SMILES length less than 128 tokens, and elemental composition limited to C, H, N, O, S, F, P, Cl, Br, and I. Only MS2 spectra corresponding to monoisotopic precursor ions were retained. For compounds with multiple adduct forms, the preferred adducts were selected in the following order: $[M+H]^+$, $[M+Na]^+$, $[M-H_2O+H]^+$ and others. Only one preferred adduct were selected per compound in the training set. The distribution of adduct types is provided in Supplementary Table 3. Each spectrum was normalized to unit intensity and denoised using an S/N threshold of 3. In total, 338,159 spectra were retained. Spectra acquired at different collision energies were merged following the previously reported method⁴¹. In brief, peaks within a 0.01 Da window were merged, and their intensities were summed. The merged spectra were then normalized again. Spectra from the same compound across different energies were then merged, and a single representative spectrum was retained for each structure. The final curated spectral dataset included a total of 23,213 unique compounds with corresponding MS2 spectra. The curated spectral dataset was partitioned into three subsets: training, validation, and testing sets. To evaluate the annotation performance on previously unseen structures, a test set comprising 1500 MS2 spectra with $[M+H]^+$ adducts was randomly selected. Both the selected spectra and their corresponding structures were excluded from the pretraining and fine-tuning datasets to ensure structure-disjoint training. The remaining spectra were randomly divided into training and validation sets at a 9:1 ratio. The training set included 19,541 merged MS2 spectra, which were used for both model training and template search. The validation set comprised 2172 merged MS2 spectra, which were used to monitor model performance and optimize the model.

To assess the structural distribution between the pretraining structure dataset and the fine-tuning spectral dataset, the maximum Tanimoto similarity was calculated for each molecule in the fine-tuning dataset against all structures in the pretraining dataset using RDKit 2048-bit molecular fingerprints. Identical structures present in the fine-tuning dataset were excluded from the pretraining dataset during the calculation. Chemical space coverage was visualized using UMAP projection ($n_neighbors = 30$, $min_dist = 0.1$). For visualization, 100,000 molecules were randomly sampled from the pretraining structure dataset to represent its overall chemical diversity. Murcko scaffold SMILES were used to calculate scaffold frequency and coverage using RDKit (v2023.9.5) (Supplementary Fig. 4).

The workflow of MetGenX

The MetGenX workflow comprises three sequential modules: template search, template encoding, and structure generation, followed by a machine learning-based re-ranking model. Detailed parameter settings for MetGenX are listed in the Supplementary Table 4.

(1) Template search. The spectral embedding approach in MetGenX was similar to previously reported strategies⁴². For each MS2 spectrum,

fragment peaks with relative intensity below 1% were removed, and up to the top 100 most intense peaks were retained. Neutral losses (NLs) between the precursor ion and each fragment were calculated, with NLs constrained to under 200 Da. Both fragment peaks and NLs were converted into textual tokens and embedded using a Word2Vec model²⁶ (Supplementary Fig. 1a). For example, a fragment with m/z 68.05 Da was converted to “peak@68.05” and a neutral loss with 18.01 Da was converted to “nl@18.01”. The Word2Vec model was unsupervised trained in 338,159 spectra from the NIST20 spectral database. The training parameters were as follows: $sg = 0$ (CBOW), $negative = 5$, $vector_size = 512$, $window = 600$, $min_count = 10$, $alpha = 0.025$, $min_alpha = 0.0125$ and $epochs = 50$. The model with the lowest training loss was selected for downstream use. After embedding, each spectrum was converted into a matrix of vectors with shape (N_p , 512). N_p denotes the number of peaks. This matrix was then pooled into spectral vector through weighted mean of intensity:

$$W_s = \sum W_t \cdot I^{0.5} \quad (1)$$

Whereas W_s is the spectral vector, W_t represents the embedded vector of each peak, and I denotes the corresponding normalized intensity.

Each spectrum in the spectral dataset was embedded into a fixed-length spectral vector. To enable fast template search, the embedded vectors were indexed using Faiss (v1.8.0), and cosine similarity was used to compute pairwise spectral similarity. For the generation model, the number of retrieved templates was limited to a maximum of 10 per spectrum to reduce computational load. Additionally, a similarity threshold of 0.4 was applied, under which 92.0% of spectra in the validation set were able to retrieve at least one template (Supplementary Fig. 1f).

To identify the most suitable method for calculating spectral similarity, we evaluated five alternative search approaches alongside our Word2Vec method (Supplementary Fig. 1b–e): modified cosine score⁸ (as implemented in GNPS), DREAMS⁴³ (dreams v1.0.0), spectral entropy⁴⁴ (ms-entropy v1.3.3), MSBert⁴⁵ (msbert v1.0), and MS2DeepScore⁴⁶ (ms2deepscore v2.0.0). Detailed parameter settings for each method are provided in Supplementary Table 5. Among these methods, the Word2Vec model was ultimately selected for its rapid performance in both spectral embedding and similarity search, while maintaining comparable structural similarity between the target structures and retrieved templates.

(2) Template encoding. Template structures were encoded using 3878-bit molecular fingerprints defined by SIRIUS (v 5.8.1), which have demonstrated the capacity to reflect the relationship between structures and MS2 spectra³⁷. The molecular formulae of templates were encoded as 10-bit vectors based on the counts of elements. Formula differences between the templates and the query spectrum were also encoded as 10-bit vectors. All vectors were concatenated to form the final feature representation for input to the generative model (Supplementary Fig. 2). Although MetGenX was trained on ten common biological elements (C, H, N, O, S, F, P, Cl, Br, I), the framework is modular and can be extended to additional element types when sufficient spectral data become available.

(3) Structure generation. The deep generative model is built upon a transformer-based sequence-to-sequence architecture (Supplementary Fig. 3a). The extracted template features were first transformed into a 512-dim matrix using a three-layer multilayer perceptron (MLP) module, which was pooled to extract high-level structural representations. The encoded structural representations were subsequently fed into a 6-layer transformer decoder cell, which proceeded autoregressively and generated the output SMILES sequence token by token. At each decoding step, the hidden states of the decoder were transformed into logits over the vocabulary via a language modeling

head. These logits were then converted to probability distributions using a softmax function.

For molecular structure generation, SMILES strings were used to represent chemical structures. A character-level tokenization strategy was applied, in which each SMILES string was decomposed into atomic tokens (e.g., “C”, “O”, “N”) and structural symbols (e.g., “(”, “#”, “)”). Notably, implicit hydrogen counts were annotated for each atom token for better formula constraint. For example, “C@3” indicates a methyl group. All tokens were mapped to a 512-dimensional vector space using a trainable embedding layer.

For database-restricted generation, the search space was restricted to candidate structures retrieved from a background database (containing 0.52 million metabolites and natural products). The next token was generated based on the given candidates, ensuring that each token in the output sequence was conditioned on candidate structures. For database-free generation, the next token was generated conditioned on the previously generated tokens. Generation was terminated when either the end-of-sequence token <e> was produced or the sequence reached the maximum length of 128 tokens. At each step, the decoder produced a probability distribution over possible next token. A formula-constrained beam search algorithm was applied with a beam width of $k = 128$ to sample top- k candidate SMILES. To avoid unnecessary expansion of the search space, the formula constraint was employed to verify formulae at each decoding step. Tokens beyond the allowed elemental composition were blocked from further expansion (Supplementary Fig. 3b). After generation, invalid SMILES strings or those not matching the target formula were discarded. For each valid candidate, the sum of the probabilities across the sequence was calculated, normalized and used as the model score.

(4) Re-ranking model. The generated candidates were further re-ranked using a lightGBM-based model (Fig. 2b). For each candidate, structural similarities between the candidate structure and each of the retrieved templates were calculated. To describe the similarity, the maximum, mean, and minimum values of structural similarities were extracted as features. Similarly, the maximum, mean and minimum value of spectral similarities between the query spectrum and the spectra of each template were calculated. Additional features included the natural product (NP) score³⁸, molecular complexity⁴⁷, SMILES length, template number, candidate rank and model score of each candidate. These features were combined to form the input feature vector for the re-ranking model. The model output candidate scores ranging from negative to positive infinity, with a higher candidate score indicating higher confidence in the ranking (Fig. 2c).

To train the re-ranking model, we utilized candidates generated from the training and validation sets. The testing set was reserved exclusively for evaluating the final performance of both the generative and re-ranking models to ensure no data leakage during evaluation. Specifically, 2000 spectra were randomly sampled from the training set to simulate targeted structural generation. During this process, templates were searched from the training set without excluding identical structures to ensure adequate diversity. For the templates with an identical structure to the targeted structure, the spectral similarity values were randomly assigned in the range of 0.7 to 1.0 to avoid overfitting. This procedure simulates experimental variability and enables the re-ranking model to learn a more realistic and robust scoring behavior, rather than overfitting to idealized similarity values. Altogether, 3995 spectra were collected for training the re-ranking model. The re-ranking model was trained to prioritize candidates with higher structural similarity using the candidate scores (S). To interpret the importance of each feature to the re-ranking model, we calculated both split counts using the LightGBM package (v4.5.0) and SHAP values using the SHAP package (v0.46.0) in Python (Supplementary Fig. 4b).

The training of MetGenX

The deep generative model of MetGenX was trained on 4 NVIDIA GeForce RTX 4090 GPUs (CUDA version 12.6). The MetGenX employed a two-step training approach: pretraining on a structure dataset and fine-tuning on a spectral dataset (Fig. 2a). In the pretraining stage, the model was trained using the curated structure dataset containing 2.17 million molecules. For each target structure, structure templates were first retrieved from the background database ($n = 0.52$ million compounds) based on structural similarity, computed using RDKit 2048-bit molecular fingerprints with a Tanimoto similarity threshold of 0.4. To avoid data leakage and overfitting, the identical structures to the target structures were not used as templates during template search. The templates were used as input to generate the corresponding target structure in a “structure-to-structure” generation task. Model training was guided by early stopping, which halted training if validation loss did not improve for 5 consecutive epochs. The pretrained model converged after 33 epochs training.

In the fine-tuning stage, the pretrained model was further optimized for metabolite generation using a curated MS2 spectral dataset. For each MS2 spectrum, templates were retrieved from the training set based on a spectral similarity threshold of 0.4 and used to guide generation in a spectrum-to-structure task. Similarly, templates with identical structures to the target structures were excluded. Early stopping was applied if the validation loss did not improve for 5 consecutive epochs. The fine-tuned model converged after 4 epochs training.

Both training stages used the AdamW optimizer with a weight decay of 1×10^{-6} . The learning rate was set to 0.01 with a linear warmup of 400 steps during pretraining, and to 5×10^{-5} during fine-tuning. The batch size was consistently set to 32 for both stages. Detailed parameter settings for model training are listed in Supplementary Table 4. The cross-entropy loss was used as the loss function and calculated as follows:

$$L = - \sum p(x) \bullet \log q(x) \quad (2)$$

where $p(x)$ represents the encoding of the correct token, and $q(x)$ denotes the predicted probability distribution.

To evaluate the contribution of key components in the deep generative model, we conducted a series of ablation experiments using the fine-tuned model as the baseline. In the ablation setting, we individually removed or modified core components to evaluate their impact on model performance (Supplementary Fig. 5a–c). First, to examine the effect of pretraining, the model was trained directly on the spectral dataset without weight initialization from the pretrained structure model (pretraining). Second, to assess the importance of template search, retrieved templates were replaced with a random selection of ten structures from the reference spectral database during model training (template search). Third, in database-free generation, the formula constraint was removed in order to evaluate the role of formula-constrained decoding (formula constraint). All ablation models were trained under the same conditions as the baseline model, ensuring that observed performance changes were directly attributable to the removal of each specific component.

Evaluation metrics

For metabolite annotation, Top-N accuracy was used to evaluate ranking performance, defined as the proportion of cases in which the correct structure appears within the top-N generated candidates. Structural correctness was determined by matching the first layer of the InChIKey between each generated structure and the target structure. In addition to exact correct matches, structural similarity was also evaluated by calculating the Tanimoto similarity of 2048-bit RDKit molecular fingerprints between the generated structures and the target structure. Furthermore, the relative ranking position (RRP)

employed to describe the relative position of the truly annotated structure across structures³².

$$RRP = \frac{1}{2} \left(1 - \frac{BC - WC}{TC - 1} \right) \quad (3)$$

whereas BC represents the number of structures that have a higher score than the target structure, and WC represents the number of structures that have a lower score. TC represent to the number of generated structures. RRP of 1.0 means the target structure ranked first, RRP of 0 means the target structure ranked last. Higher RRP means better ranking performance for annotation.

Biological sample preparation and data acquisition

A total of five biological sample types were analyzed: BV2 cells ($n = 6$; biological replicates), mouse brain tissue ($n = 6$; biological replicates), mouse liver tissue ($n = 6$; biological replicates), NIST human plasma ($n = 6$; technical replicates), and NIST human urine ($n = 6$; technical replicates). All sample preparation followed common protocols. The NIST human urine (SRM 3667) and NIST human plasma (SRM 1950) reference materials were purchased from Ango Biotechnology Co. (Shanghai, China). Human plasma or urine samples (100 μ L) were extracted using MeOH (300 μ L). After vortexing for 30 s, the samples were subjected to sonication for 15 min and then incubated for 1 h at -20°C , followed by centrifugation at $17,000\times g$ for 15 min at 4°C to precipitate proteins. The resulting supernatants were collected, transferred into LC sample vials and stored at -80°C prior to LC-MS analysis. The 6-week-old male C57BL/6J mice ($n = 6$) were group-housed in a barrier facility under controlled conditions (at room temperature of 22°C with 50% humidity and 12 h light/12 h dark cycles). The C57BL/6J strain was obtained from Vital River Laboratories (Beijing, China). For metabolite extraction of mouse tissues, 20 mg of tissue was homogenized in 200 μ L of H_2O under low temperature conditions, continuously cooled using N_2 gas flow from liquid nitrogen. Subsequently, 800 μ L of MeOH:ACN (1:1, v/v) was added to 200 μ L of the homogenate. The mixture was vortexed for 30 s and sonicated for 10 min in a 4°C water bath. The samples were incubated at -20°C for 1 h to facilitate protein precipitation, followed by centrifugation at $17,000\times g$ for 15 min at 4°C . The supernatant was collected and evaporated to dryness at 4°C using a vacuum concentrator. Dried samples were stored at -80°C prior to LC-MS analysis. BV2 cells were plated in 6-cm dishes at 2×10^6 cells per dish and cultured in DMEM medium supplemented with 10% FBS and 1% penicillin/streptomycin. The extraction solution ACN/MeOH/ H_2O (2:2:1, v/v/v) was pre-cooled at -80°C for 1 h before use to ensure no ice formation. When cells reached the target density, the culture medium was completely aspirated, and the cells were washed rapidly with 1 mL of 37°C PBS. The dish was placed on dry ice, and 1000 μ L of pre-cooled extraction solution was added, followed by incubation at -80°C for at least 40 min. During transfer, the dish was kept on dry ice. Cells were then scraped and transferred into a 2-mL EP tube. Then an additional 500 μ L of extraction solution was used to rinse the dish. The combined extract (1.5 mL total) was collected. The samples were vortexed for 1 min at 4°C and centrifuged at $17,000\times g$ for 15 min at 4°C to pellet insoluble materials. The resulting supernatant was transferred to a new EP tube and evaporated to dryness at 4°C using a vacuum concentrator. Dried samples were stored at -80°C until analysis. Prior to LC-MS analysis, dried extracts of mouse tissue and BV2 cells were reconstituted in 100 μ L of ACN: H_2O (1:1, v/v), vortexed for 30 s, and sonicated for 10 min at 4°C in a water bath. The reconstituted samples were then centrifuged at $17,000\times g$ for 15 min at 4°C , and the resulting supernatant was transferred into LC sample vials for analysis.

The untargeted metabolomics data were acquired by LC-MS using a Vanquish UHPLC system coupled to an Orbitrap mass spectrometer (Exploris 480, Thermo Fisher Scientific). For LC separation, a

Waters ACQUITY UPLC BEH Amide column (1.7 μ m particle size, 100 mm length $\times$ 2.1 mm inner diameter) was used for HILIC, and a Kinetex C18 column (2.6 μ m, 2.1×100 mm) was used for reverse phase (RP) separation. The column temperature was maintained at 25°C , and a 2 μ L injection volume was commonly used. For HILIC separation, the mobile phase A consisted of 25 mM ammonium hydroxide (NH_4OH) and 25 mM ammonium acetate (NH_4OAc) in water, and B consisted of acetonitrile (ACN), in both the positive (ESI+) and negative (ESI-) modes. The flow rate was 0.5 mL/min, and the gradient was programmed as follows: 0–0.5 min, 95% B; 0.5–7 min, 95% B to 65% B; 7–8 min, 65% B to 40% B; 8–9 min, 40% B; 9–9.1 min, 40% B to 95% B; and 9.1–12 min, 95% B. For RP separation, mobile phase A was 0.01% acetic acid in water, and B was a mixture of isopropyl alcohol and acetonitrile (1:1), in both the positive (ESI+) and negative (ESI-) modes. The flow rate was 0.3 mL/min, and the gradient was programmed as follows: 0–1 min: 1% B; 1–8 min: 99% B; 8–9 min: 99% B; 9.0–9.1 min, 99% B to 1% B; and 9.1–12 min: 1% B. The LC-MS data were acquired in full scan mode with positive/negative ion polarity switch, and ddMS2 scans were applied to obtain MS2 spectra for all samples. The ESI source parameters were as follows: spray voltage, 3500 V (positive) or -2800 V (negative), respectively; sheath gas, 50 arbs; aux gas, 15 arbs; sweep gas, 2 arbs; ion transfer tube temperature, 350°C ; vaporizer temperature, 400°C for HILIC or 350°C for RPLC analysis. Full MS scan was set as follows: orbitrap resolution, 60,000; AGC target, 1e6; maximum injection time, 100 ms; scan range, 70–1200 Da. The ddMS2 scan was set as: orbitrap resolution, 30,000; AGC target, 1e5; maximum injection time, 60 ms; scan range, 50–1200 Da; top N, 6; isolation width, 1.0 m/z ; collision energy mode, stepped; collision energy type, normalized; HCD collision energies (%), 20–30–40; exclusion duration 4 s.

Raw LC-MS data files (.raw) were converted to.mzXML format for MS1 and.mgf format for MS2. XCMS (v 3.12.0) was used for peak detection, retention time correction, and peak alignment. The key parameters were set as follows: Method: centWave; Peak width: min = 5, max = 30; S/N threshold: 3; Minimum fraction: 0.5. The metabolites in biological samples were annotated using MetDNA3³⁶ (v 3.1). The resulting peak tables and MS2 spectra were matched against an in-house MS2 spectral library and retention time (RT) library to achieve level 1 identification results. The in-house library consisted of RT and MS/MS spectra acquired from purchased chemical standards. The metabolite annotation was employed in identical experimental conditions between biological sample and in-house library. Matching tolerances were defined as follows: MS1 match, 15 ppm; Retention time, 25 s; MS2 match: dot product score > 0.8.

Annotation of metabolites in real biological samples

To evaluate the performance of MetGenX on real biological samples, we first annotated MS2 spectra from five different sample types using library-based spectral matching. Annotation results were manually checked to obtain level 1 identifications. Only metabolites present in the background database were retained for validation. In total, 1720 MS2 spectra were identified across these samples. MetGenX was subsequently applied to annotate these spectra in database-restricted mode, using a background database containing 0.52 million chemical structures. To ensure that template search matched the experimental conditions of biological samples, reference spectra were curated from the NIST database by selecting, for each compound, the MS2 spectrum acquired at a collision energy closest to normalized collision energy (NCE) 30%. This curated subset was used as the reference spectral database for template search, with a cosine similarity threshold of 0.4. Using this approach, 1681 out of 1720 spectra (97.7%) successfully retrieved templates from the full NIST spectral dataset for generation. Annotation performance was evaluated using annotation coverage and top-N accuracy. Annotation coverage was defined as the number of MS2 spectra for which at least one candidate structure was

successfully generated. Top-N accuracy was calculated as the proportion of correctly annotated spectra among these successfully annotated cases. For spectra with multiple matched annotation results, the best rank was used to compute the top-N accuracy.

To benchmark annotation accuracy, MetGenX was compared with five widely used in silico MS2 annotation tools: CFM-ID¹¹ (v 4.0), MetFrag³⁴ (v 2.4.6-CL), MS-FINDER¹² (v 3.6.1), ICEBERG³⁵ (<https://github.com/coleygroup/ms-pred>), and MSNovelist¹⁵ (incorporated in SIRIUS v 6.0.6). To ensure a fair and consistent comparison, all methods were provided with the same background database as used by MetGenX. The correct molecular formula was assumed to be known for each MS2 spectrum. ICEBERG was retrained on the same dataset following the provided training protocols, whereas other tools were used with their embedded pretrained models. Detailed parameter settings for each tool are listed in Supplementary Table 6.

Metabolite generation in negative ionization mode

MS2 spectra acquired in negative ionization mode using Orbitrap instruments were collected from the NIST20 spectral database and cleaned as for the positive mode. A Word2Vec model was trained on 129,489 MS2 spectra in negative mode, using the same parameters as those applied in the positive mode. The generative model and re-ranking model trained in the positive mode were directly applied for structure generation. For performance evaluation in the NIST spectral database, shared metabolites between positive and negative ionization modes were identified from the testing set in positive mode. Subsequently, MS2 spectra acquired at different collision energies in negative mode for each compound were merged to construct the testing set, including 653 merged MS2 spectra. Templates were retrieved from a reference spectral database of 7679 merged MS2 spectra. This setup ensured no structural overlap during the template search and generation process. The same configuration and background database as in the positive mode were applied for both database-restricted and database-free generation.

For metabolite annotation in biological samples, a total of 2417 MS2 spectra from biological datasets were used for evaluation. The same configuration and background database as in the positive mode were applied during structure generation. Templates were retrieved from the full NIST spectral dataset comprising 9826 merged spectra acquired at a collision energy closest to NCE 30%, using a cosine similarity threshold of 0.4. In database-restricted mode, MetGenX successfully annotated 2319 spectra. Comparison with other in silico annotation tools was conducted using the same procedures as in the positive mode.

Spike-in experiments

To evaluate the annotation performance of MetGenX in real biological samples, 200 metabolite standards were spiked into mouse liver and NIST 1950 plasma samples. The list of spiked metabolites followed our previously reported study⁴⁸, and detailed compound information was provided in Supplementary Data 12. Untargeted metabolomics analyses were performed under the same LC–MS conditions as those used for biological samples. The spike-in metabolites were identified by matching *m/z* values, retention times (RT), and MS2 spectra with those of the corresponding chemical standards to obtain level 1 annotation. For MetGenX, annotation accuracy was calculated across all detected metabolites, using the minimum annotation rank obtained from different LC–MS methods to eliminate duplicated metabolites. For other annotation tools, the parameters were kept identical to those listed in Supplementary Table 6 to annotate the corresponding experimental MS2 spectra of the spiked metabolites in both positive and negative ionization modes.

Comparison with the network-based annotation method

MetGenX was compared with MetDNA3 across 20 biological sample datasets as described previously³⁶. For MetDNA3, 30% of level 1

annotated metabolites were randomly selected as seed compounds to initiate network-based annotation propagation along the metabolic reaction network (MRN), while the remaining 70% were reserved for validation under a 10-fold cross-validation approach. For MetGenX, the database-restricted mode was applied using the same MRN (175,354 metabolites) used by MetDNA3, which served as the background database for a fair comparison. All annotated structures were excluded during template search for each dataset. RT filtering was performed using predicted RT values with a 30% tolerance, based on the same random forest model as MetDNA3. Annotation performance was evaluated using coverage and top-3 accuracy, following the same definitions and procedures as in this study and the MetDNA3 work³⁶.

Discovery of metabolites in liver sample

The metabolomics dataset from mouse liver included 6564 metabolic features with corresponding MS2 spectra acquired in positive ionization mode. Molecular formulae were initially predicted using SIRIUS³⁷ (v 6.0.6). The MS2 mass accuracy was set to 10 ppm, and adduct types were limited to $[M+H]^+$ and $[M+Na]^+$. To ensure high-confidence formula annotations, only results meeting the following criteria were retained: *m/z* error < 10 ppm, explained intensity > 0.8, and Zodiac score ≥ 0.1 , resulting in 2979 features with confident formula assignments.

A multi-step annotation workflow was then applied (Fig. 6a). First, MS2 spectra were matched against an in-house MS2 and retention time library using previously described parameters, yielding 516 level 1 annotations. The remaining spectra were annotated using MetGenX. During structure generation, templates were retrieved from the full NIST dataset acquired at NCE30, using a cosine similarity threshold of 0.4. For high-quality candidate selection, a candidate score threshold (*S*) of 1.2 was applied. This threshold was empirically determined from metabolites identified in the mouse liver dataset, achieving a true positive rate of 0.80 and a false positive rate of 0.12 for distinguishing correct from incorrect annotations (Supplementary Fig. 10a, b). In database-restricted generation, structure candidates were retrieved from the background database (0.52 million structures) based on the predicted molecular formula, resulting in 541 spectra with high-confidence candidates (*S* > 1.2). The remaining spectra were annotated using database-free generation, resulting in 426 spectra with high-quality candidates and 1102 spectra with at least one valid structure.

The adenosyl-cysteine is validated with chemical standards through chemical syntheses, as previous reported route⁴⁹. The Ap2I is validated with chemical standards through chemical syntheses as below: AMP (347 mg, 1 mmol) was dissolved in 10 mL of *t*-BuOH, followed by dropwise addition of morpholine (5 eq.). The reaction mixture was first stirred at room temperature for 15 min and then set to reflux. Dicyclohexylcarbodiimide (DCC, 824 mg, 4 eq.) was dissolved in 10 mL of *t*-BuOH and was added dropwise to the reaction mixture under reflux every 10 min. After 2 h, DCC addition was complete, and reflux was continued for another 5 min. After the reaction was completed, the mixture was cooled to rt and volatiles were removed under vacuum. The residue was treated with 2 mL of water. Solids were removed by filtration, and the aqueous layer was concentrated under vacuum to yield the crude product. The crude product was dissolved in water, and automated flash column chromatography was performed using a Teledyne ISCO CombiFlash Companion system with a RediSepRf reverse-phase C18 column, packed with 40 g silicagel by using 100 mM NH_4HCO_3 : acetonitrile (98%:2% to 85%:15%) as an eluent with a flow rate of 40 mL/min to get 150 mg AMP-morpholine. The 5 mL tube was charged with AMP-morpholine (0.05 mmol, 1 eq.). The coupling IMP (0.05 mmol, 1 eq.) was added, along with $MgCl_2 \cdot (H_2O)_6$ (0.076 mmol, 15.2 mg, 1.5 eq.), tetrazole (0.1 mmol, 7 mg, 2 eq.), H_2O (0.3 mmol, 31.2 μ L, 6 eq.) and a 50 mm stainless steel ball. The ball mill was set to vibrate at a frequency of 30 Hz for 90 min, after which time the jar was left to cool down to room temperature. Once the jar was

opened, the product was dissolved in water, and purified by prep-HPLC by using 100 mM formic acid: acetonitrile (98%:2% to 85%:15%) as an eluent with a flow rate of 10 mL/min to get 5 mg Ap2I product. The structure was confirmed by nuclear magnetic resonance spectroscopy (^1H NMR, 400 MHz, D_2O) δ (ppm) = 8.13 (d, J = 24 Hz, 1H), 8.10 (s, 1H), 7.94 (d, J = 16 Hz, 1H), 7.90 (s, 1H), 5.84 (d, J = 12 Hz, 2H), 4.48–4.43 (m, 2H), 4.33–4.32 (m, 2H), 4.25–4.19 (m, 4H), 4.13–4.09 (m, 2H).

Benchmarking in MassSpecGym datasets

MetGenX was evaluated on the MassSpecGym dataset following the training-testing split for fair comparison with previously reported baselines³⁹. The model was initialized with weights pretrained on a structure dataset and subsequently fine-tuned on MassSpecGym within the MetGenX framework (Fig. 2a, b). During pretraining, MetGenX was trained on compounds from HMDB²⁸, DSSTOX⁵⁰, COCONUT²⁷, and MOSES²⁴, as curated by DiffMS²¹. To prevent data leakage, all molecules overlapping with the MassSpecGym validation and test sets were explicitly removed. Structure templates were retrieved from the MassSpecGym training set based on structural similarity using a Tanimoto similarity threshold of 0.4. During fine-tuning, template search was also restricted to spectra in the training set with a spectral similarity threshold of 0.4. Spectra collected under different collision energies, instrument types, and adduct forms were treated as independent entries and were not merged during model training or evaluation. Duplicate templates with identical structures were removed by retaining only the search result with the highest score, and up to 10 templates were retained for model training and structure generation. The re-ranking model was not incorporated during benchmarking to reduce computational overhead and ensure comparable evaluation. In addition, we evaluated a MetGenX-Full model where the similarity threshold in the template search of MetGenX was removed and ten structure templates were selected with the highest similarity scores as inputs for MetGenX, enabling full-spectrum coverage across the dataset. For both training stages, the batch size was set to 64, and the learning rate was adjusted to 5×10^{-6} using the Adam optimizer (weight decay = 1×10^{-6}) to accommodate different dataset scales. All other hyperparameters were kept identical to those listed in Supplementary Table 4. Model performance was evaluated using the standard MassSpecGym benchmark metrics.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Unless otherwise stated, all data supporting the results of this study can be found in the article, supplementary, and source data files. The NIST20 database used in this study is a commercial resource and cannot be publicly redistributed. Access can be obtained by purchasing the NIST Mass Spectral Library from NIST. The structure generation and metabolite annotation results are provided in Supplementary Data 1–6. All MS2 spectral datasets used to evaluate MetGenX are provided in Supplementary Data 7–10, including the compound information from the NIST testing set and MS2 spectra from biological samples. MS2 spectra from the mouse liver metabolomics dataset are provided in Supplementary Data 11. Detailed compound information and annotation results of spike-in experiments are provided in Supplementary Data 12. The raw data files of BV2 cells, mouse brain tissue, mouse liver tissue, NIST human plasma, and NIST human urine from the Orbitrap instrument can be accessed at the MassIVE repository [<https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?accession=MSV000097913>]. The raw data files of spike-in experiments can be accessed at the MassIVE repository [<https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?accession=MSV000101078>]. Source data are provided with this paper.

Code availability

MetGenX is written in the Python language (v 3.12.6). The source code of MetGenX is freely available on GitHub [<https://github.com/ZhuMetLab/MetGenX>] under the CC BY-NC-ND 4.0 License. The model weights and database are available in Zenodo [<https://doi.org/10.5281/zenodo.18907370>].

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Author contributions

Z.-J.Z. and H.W. conceived the idea and designed the algorithm and software. H.W. contributed to the model training, evaluation and developed the MetGenX package. H.W. and H.Z. performed the sample preparation, data acquisition and processing for biological samples. H.W. and Z.-J.Z. wrote the manuscript. Z.-J.Z. supervised the project. We thank Dr. Nan Liu and Jinxin Li for their assistance in synthesizing the chemical standards.

Competing interests

The authors declare no competing interests.

Additional information

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